THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 7: **Temperature and its Measurement** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why is it risky to put a mercury thermometer in the oven? 1

(a) it becomes plastic **(b)** it is cold

(c) it can break **(d)** it will show correct reading

2. Which of the following cannot be measured by a thermometer? 1

(a) mass **(b)** temperature of a room

(c) temperature of milk **(d)** temperature of a body

3. A hot spoon and a cold glass are placed together. The final temperature of the spoon will: 1

(a) become 100 °C **(b)** rise above the glass

(c) fall **(d)** stay constant

4. Which of the following is the best heat insulator? 1

(a) aluminium **(b)** air **(c)** iron **(d)** copper

5. A student tabulates the temperature of water every minute as it cools: 90, 80, 72, 66, 61, 57. The cooling is: 1

(a) faster at first then slower **(b)** stops

(c) increases **(d)** constant

6. Which of these will conduct heat the fastest? 1

(a) wooden rod **(b)** copper rod **(c)** glass rod **(d)** plastic rod

7. A woollen blanket keeps us warm in winter mainly because: 1

(a) wool is hot **(b)** trapped air reduces heat loss

(c) wool is a metal **(d)** wool produces heat

8. The reading 36.9 °C of a child is: 1

(a) dangerous **(b)** low **(c)** normal **(d)** high fever

9. Why is mercury preferred to water as a thermometric liquid? 1

(a) it is cheap

(b) it is cold

(c) it expands uniformly and does not wet glass

(d) it is transparent

10. Why do we use two thin blankets instead of one thick one for more warmth? 1

(a) blankets produce heat **(b)** blankets are cheap

(c) easy to wash **(d)** air layer between them insulates

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why a mercury thermometer is slow to rise when a large cold block is placed with it. 2

12. Why is aluminium foil used for wrapping hot food? 2

13. Why does the temperature of a hot cup of tea fall while that of the surroundings rises slightly? 2

14. Why is a thermometer kept in the shade to read air temperature? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why water in an earthen pot (matka) remains cool in summer. 3

16. Compare the reading time of a mercury and a digital thermometer. 3

17. List the three states of water at different temperatures and what happens at 0 °C and 100 °C. 3

OR

A person measures a 38 °C temperature at morning and 39.5 °C in the evening. What would you conclude and suggest? 3

18. Why is it advised that a clinical thermometer should not be washed in hot water? 3

OR

Why does a spoon placed in a hot cup become warm at the top? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how we can show that heat flows from hot to cold using hot water in a metal can and a thermometer, with observations. 5

OR

Describe in steps the experiment to test three materials (iron, wood, plastic) as conductors using hot water. 5

20. Compare conductors and insulators and select materials for 5
(a) a saucepan,
(b) its handle,
(c) an oven glove,
(d) an electric kettle body.

OR

Why are some utensils like pressure cookers made with a thin base but a thick lid? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND